

ECE Probabilistic Systems Analysis

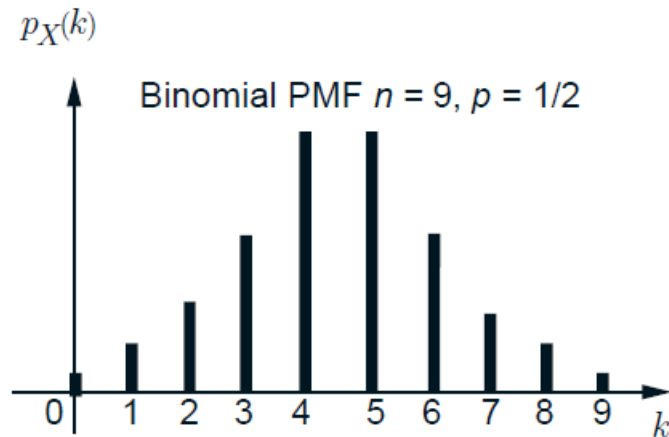
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Continuous Random Variables

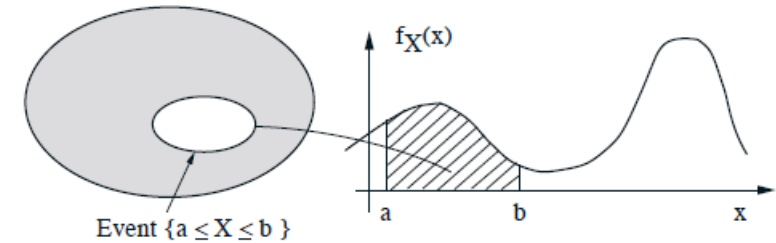
Probability Mass Function or (PMF) $p_X(x)$



$$\mathbf{P}(a \leq X \leq b) = \sum_{x: a \leq x \leq b} p_X(x)$$

$$p_X(x) \geq 0 \quad \sum_x p_X(x) = 1$$

Probability Density Function or (PDF) $f_X(x)$

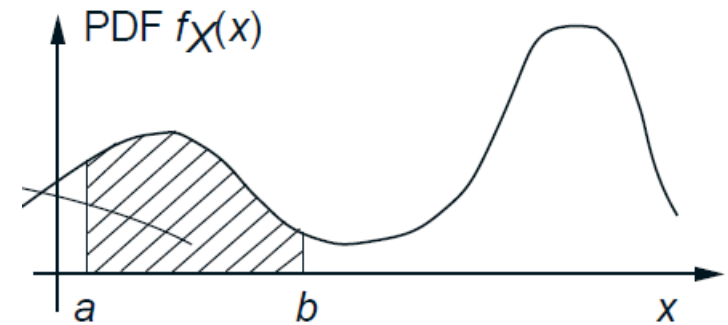


The probability that the value of X falls within an interval $[a, b]$ is:

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx,$$

$$f_X(x) \geq 0 \quad \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$f_X(x)$ is the “probability mass per unit length” near x



$$\mathbf{P}(a \leq X \leq a + \delta) \approx f_X(a) \cdot \delta$$

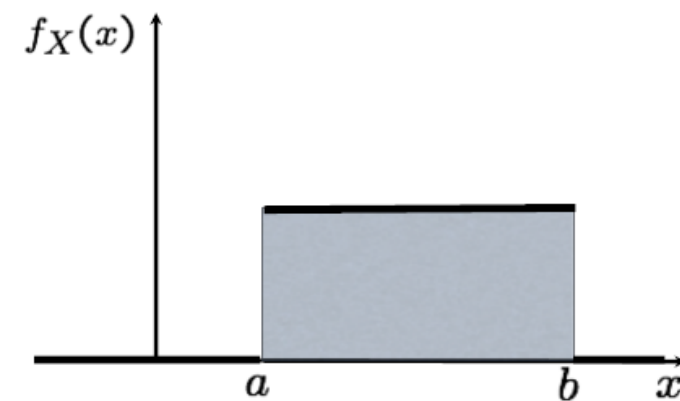
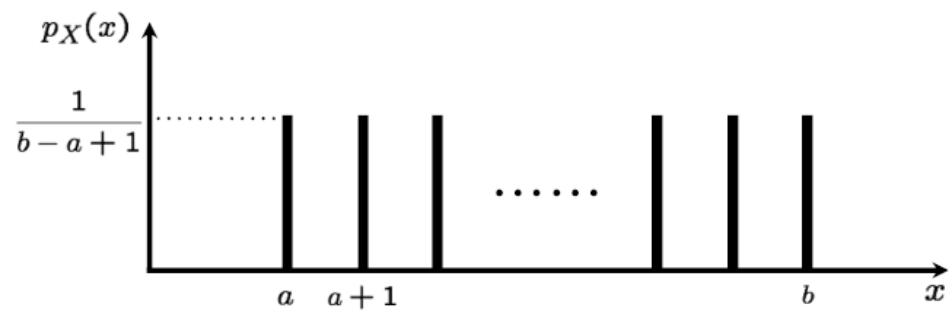
$$\mathbf{P}(X = a) = 0$$

$$\mathbf{P}(X = a) = \int_a^a f_X(x) dx = 0$$

For this reason, including or excluding the endpoints of an interval has no effect on its probability

$$\mathbf{P}(a \leq X \leq b) = \mathbf{P}(a < X < b) = \mathbf{P}(a \leq X < b) = \mathbf{P}(a < X \leq b)$$

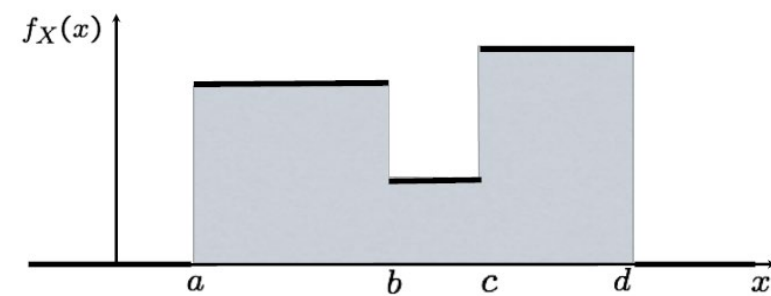
Continuous Uniform Random Variable



$$1 = \int_a^b c \, dx = c \int_a^b dx = c(b-a),$$

$$c = \frac{1}{b-a}.$$

$$f_X(x) = \begin{cases} c & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise,} \end{cases}$$

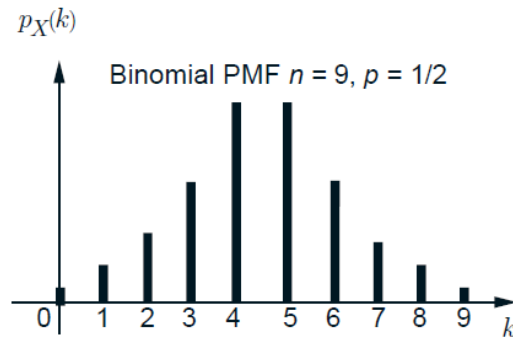


Alvin's driving time to work is between 15 and 20 minutes if the day is sunny, and between 20 and 25 minutes if the day is rainy, with all times being equally likely in each case. Assume that a day is sunny with probability $2/3$ and rainy with probability $1/3$. What is the PDF of the driving time, viewed as a random variable X ?

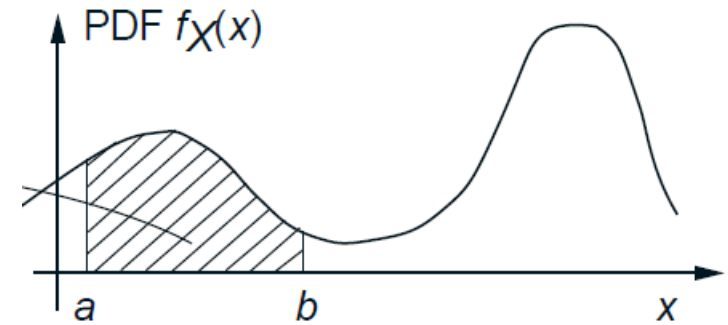
$$f_X(x) = \begin{cases} c_1 & \text{if } 15 \leq x < 20, \\ c_2 & \text{if } 20 \leq x \leq 25, \\ 0 & \text{otherwise,} \end{cases}$$

$$c_1 = \frac{2}{15}, \quad c_2 = \frac{1}{15}.$$

Expectation/ mean of a Continuous Random Variable



$$\mathbf{E}[X] = \sum_x x p_X(x)$$



$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

$\mathbf{E}[X]$ can be interpreted as the “center of gravity” of the probability law and, also, as the anticipated average value of X in a large number of independent repetitions of the experiment.

Mean and Variance

The expectation of X is defined by

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

The variance of X is defined by

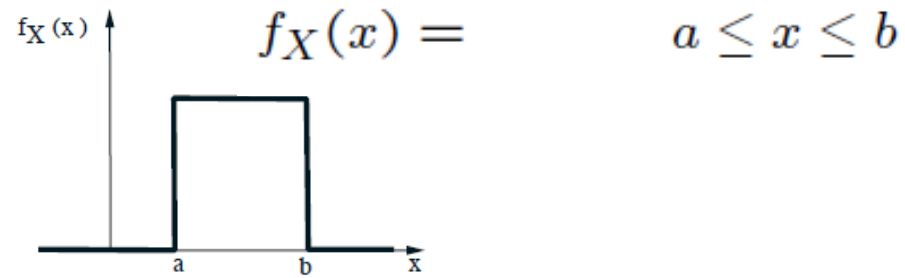
$$\text{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] = \int_{-\infty}^{\infty} (x - \mathbf{E}[X])^2 f_X(x) dx$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2.$$

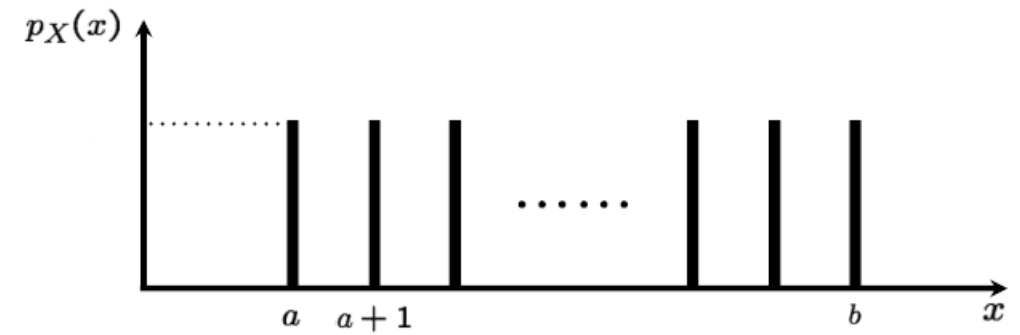
Standard Deviation

$$\sigma(x) = \sqrt{\text{var}(x)}$$

Mean and Variance of the Uniform Random Variable



$$E[X] =$$



$$E[X] = \frac{a + b}{2}$$

$$\text{var}(X) = \frac{1}{12}(b - a)(b - a + 2)$$

Mean and Variance of the Uniform Random Variable

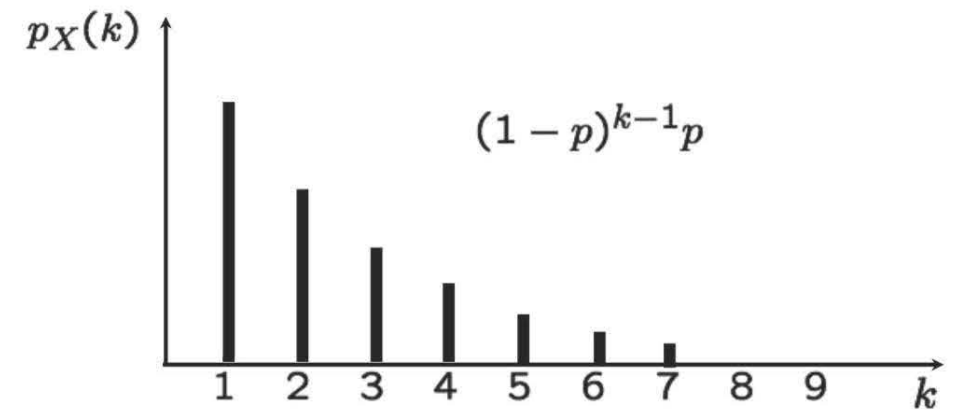
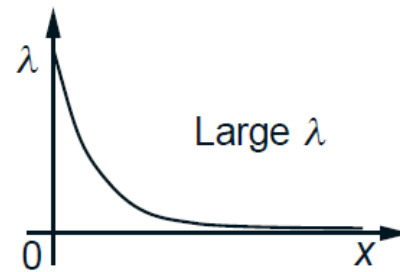
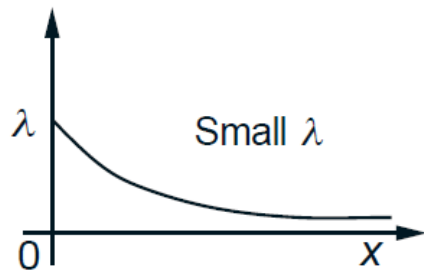
$$\mathbf{E}[X^2] =$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a + b)^2}{4} = \frac{(b - a)^2}{12},$$

$$\sigma_X^2 = \int_a^b \left(x - \frac{a + b}{2}\right)^2 \frac{1}{b - a} dx = \frac{(b - a)^2}{12}$$

Mean and Variance of the Exponential Random Variable

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



$$\mathbf{E}[X] =$$

$$\mathbf{E}[X^2] =$$

$$\text{var}(X) =$$
